

MH370 primary impact - CMEMS ocean-drift diagnostic sensitivity - conditional sensitivity

Full-domain GLORYS12 currents + distinct WAVERYS Stokes; ESS 924.6 (baseline 1930.3); not a central posterior update

Estimate at a glance

BRANCH

ocean-cmems-glorys12-waverys-diagnostic

SURFACE FAMILY

cmems-glorys12-waverys-full-domain

SUPPORTED PARENT MASS

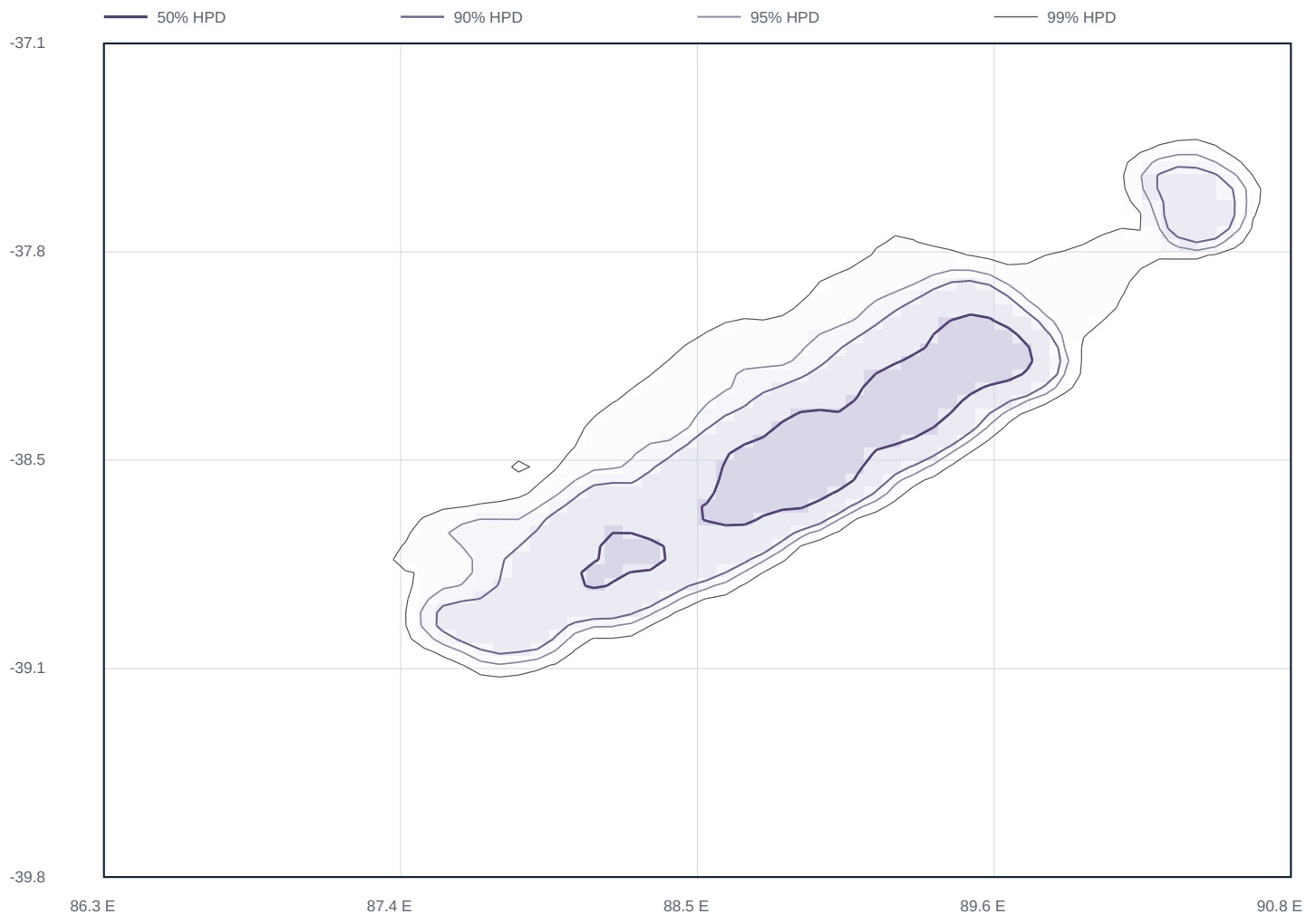
100.000%

MEAN SHIFT

26.89 NM

ESS BASELINE / CONDITIONAL

1930.3 / 924.6



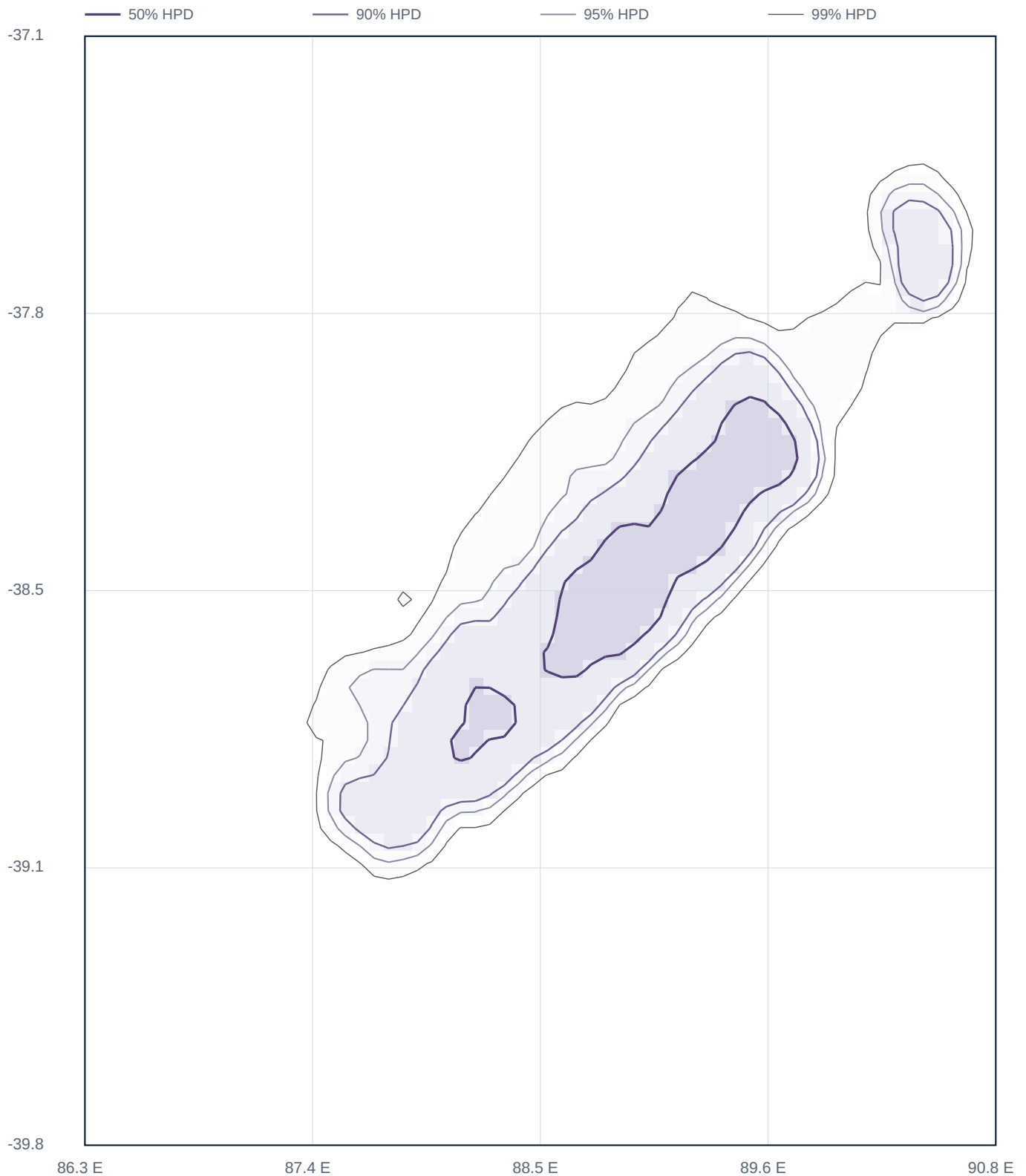
Probability is conditional on the named motion, SATCOM, and evidence configuration.

HPD contours use a fixed 5-cell display kernel; estimator weights and numerical summaries are unchanged.

Posterior geography

Highest-density regions are computed from the weighted particle population.

HPD contours use a fixed 5-cell display kernel; estimator weights and numerical summaries are unchanged.



Interpretation

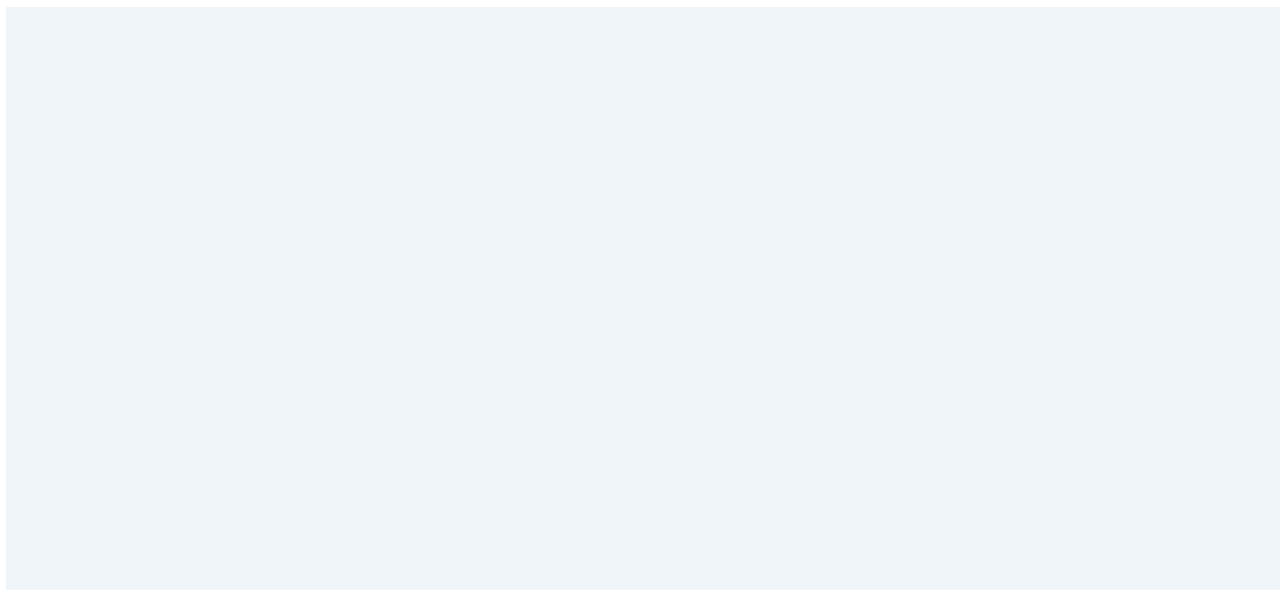
The map is a probability density over the modelled impact state. It is not a searched-area elimination map, and structural model alternatives must be compared separately.

Numerical behaviour and conditions

Adaptive likelihood tempering, particle diversity, and declared evidence conditions.

Effective sample size

peak 1



Evidence conditions

- Application status: provisional_conditional_sensitivity
- Support policy: condition_on_native_support
- Central posterior updated: false

Material limitations

- GLORYS12V1 assimilates ocean observations; the drifter replay is not guaranteed statistically held out
- WAVERYS assimilates significant wave height and uses GLORYS12 currents in its wave calculation
- one-times surface Stokes response, object leeway, diffusivity, recovery kernels, and beaching-to-discovery delays are sensitivities rather than jointly out-of-sample calibrated parameters
- the persistent GLORYS12/WAVERYS support-mask union distinguishes land encounter but is not a measured shoreline or calibrated beaching, retention, or refloating model
- this family remains separate from native HYCOM and GDP; no subjective family weight or pooled surface is reported
- Family decision diagnostic_only failed: out_of_support_termination, eventwise_arrival_ess, seed_and_particle_stability.
- This release exposes the source surface only as an explicitly enabled conditional sensitivity; it does not change the central posterior.
- Out-of-support parent mass 0.000000 is removed only by the explicit 'condition_on_native_support' sensitivity condition; it is not evidence against those particles.